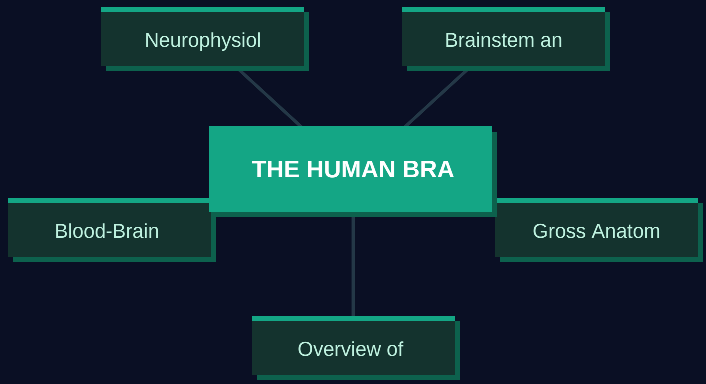
THE HUMAN BRAIN: Structure, Function, and Clinical Significance

A Comprehensive Medical Education Review

8 Slides Medical Education



Overview of the Human Brain

1

1 The adult human brain weighs approximately 1300 to 1400 grams on average.

2 Contains roughly 86 billion neurons and an equal number of glial cells.

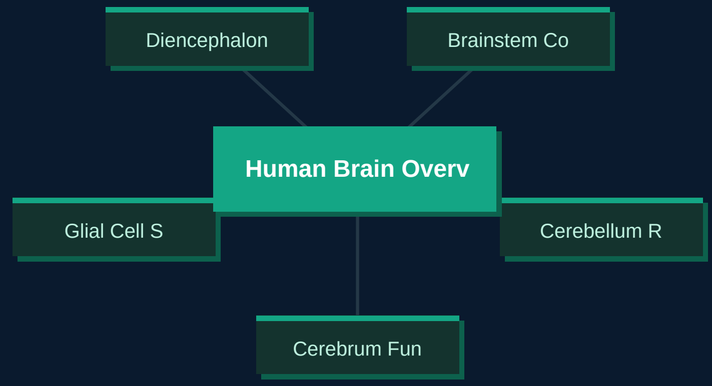
3 Consumes 20% of total body oxygen despite comprising only 2% of body weight.

4 Cerebral blood flow averages 750 mL per minute under normal resting conditions.

5 Brain is organized into cerebrum, cerebellum, brainstem, and diencephalon *regions.*

6 Gray matter contains neuronal cell bodies; white matter contains myelinated *axonal tracts.*

CONCEPT MAP



KEY INSIGHT

Brain uses 20% of O2 yet weighs only 2% of body mass.

Gross Anatomy and Major Divisions

2

1 Cerebrum divided into frontal, parietal, temporal, occipital, and insular lobes.

2 Frontal lobe governs motor control, executive function, and personality regulation.

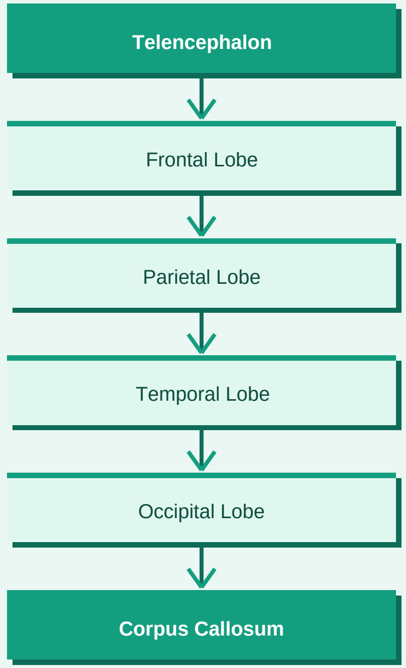
3 Parietal lobe integrates somatosensory input and spatial orientation processing.

4 Temporal lobe processes auditory information and is critical for memory formation.

5 Occipital lobe is the primary visual cortex receiving retinal projections via optic *radiations.*

6 Corpus callosum contains 200 to 250 million axons connecting bilateral cerebral *hemispheres.*

MECHANISM FLOWCHART



KEY INSIGHT

Corpus callosum has 200-250 million axons linking both hemispheres.

Brainstem and Cerebellum

1. Brainstem comprises midbrain, pons, and medulla oblongata in descending order.

2. Medulla oblongata contains vital centers controlling respiration, heart rate, and *vasomotor tone.*

3. Pons houses cranial nerve nuclei V through VIII and the pneumotaxic center.

4. Midbrain contains superior and inferior colliculi for visual and auditory reflex *pathways.*

5. Cerebellum has 3 lobes and coordinates voluntary movement, balance, and fine *motor control.*

6. Purkinje cells of cerebellum are the sole output neurons of the cerebellar cortex.

MECHANISM FLOWCHART

Midbrain Structure
↓
Pons Functions
↓
Medulla Oblongata
↓
Cerebellar Input
↓
Cerebellar Cortex
↓
Motor Coordination Output

KEY INSIGHT
Medulla controls breathing and heart rate; damage is rapidly fatal.

Neurophysiology and Neurotransmitters

1. Resting membrane potential of neurons is approximately negative 70 millivolts at *baseline.*

2. Action potential threshold is reached at approximately negative 55 millivolts in *most neurons.*

3. Dopamine pathways include mesolimbic, mesocortical, nigrostriatal, and *tuberoinfundibular systems.*

4. Serotonin modulates mood, sleep, appetite, and pain; deficiency is linked to *depression.*

5. GABA is the primary inhibitory neurotransmitter; glutamate is the primary *excitatory neurotransmitter.*

6. Acetylcholine mediates neuromuscular junction transmission and hippocampal *memory consolidation pathways.*

MECHANISM FLOWCHART

Resting Potential
↓
Action Potential Firing
↓
Neurotransmitter Release
↓
Synaptic Cleft Crossing
↓
Receptor Binding
↓
Postsynaptic Response

KEY INSIGHT

GABA inhibits; glutamate excites; imbalance drives seizures and neurodegeneration.

Blood-Brain Barrier and CSF

1. Blood-brain barrier formed by tight junctions between cerebral capillary *endothelial cells.*

2. Astrocyte endfeet surround capillaries and maintain barrier integrity and ion *homeostasis.*

3. Cerebrospinal fluid is produced by choroid plexus at approximately 500 mL per *day.*

4. Total CSF volume is 100 to 150 mL; turnover occurs approximately 3 times daily.

5. Normal CSF pressure is 7 to 18 cm H_2O in lateral recumbent position.

6. CSF flows from lateral ventricles through foramen of Monro to third and fourth *ventricles.*

MECHANISM FLOWCHART

Choroid Plexus Production
↓
Lateral Ventricles
↓
Third Ventricle
↓
Fourth Ventricle
↓
Subarachnoid Space
↓
Arachnoid Granule Reabsorp

KEY INSIGHT

CSF turns over 3 times daily; BBB excludes 98% of drugs.

Common Neurological Disorders

1. Stroke affects 15 million people worldwide annually; 5 million die each year.

2. Ischemic stroke accounts for 87% of all strokes; hemorrhagic for 13% of cases.

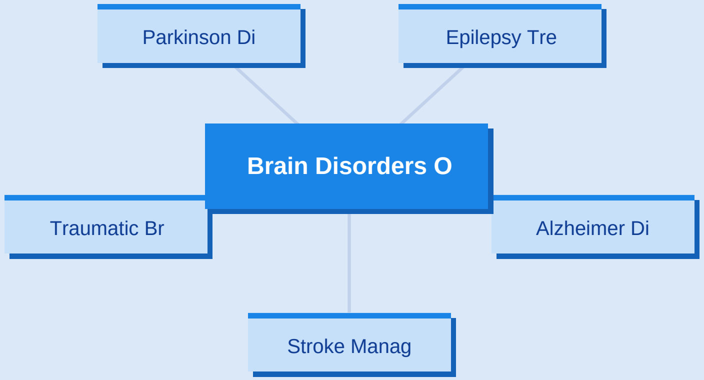
3. Alzheimer disease is the most common dementia affecting 50 million people *globally.*

4. Epilepsy affects 65 million worldwide; 70% achieve seizure control with *antiepileptic drugs.*

5. Parkinson disease involves progressive dopaminergic neuron loss in the *substantia nigra pars compacta.*

6. Traumatic brain injury causes 69 million new cases annually with significant global *morbidity.*

CONCEPT MAP



KEY INSIGHT
Ischemic stroke comprises 87% of cases; tPA within 4.5 hours saves lives.

Clinical Assessment and Management Pathways

1. Glasgow Coma Scale scores range 3 to 15; score below 8 indicates severe brain *injury.*

2. CT scan is first-line imaging for acute stroke and traumatic brain injury *assessment.*

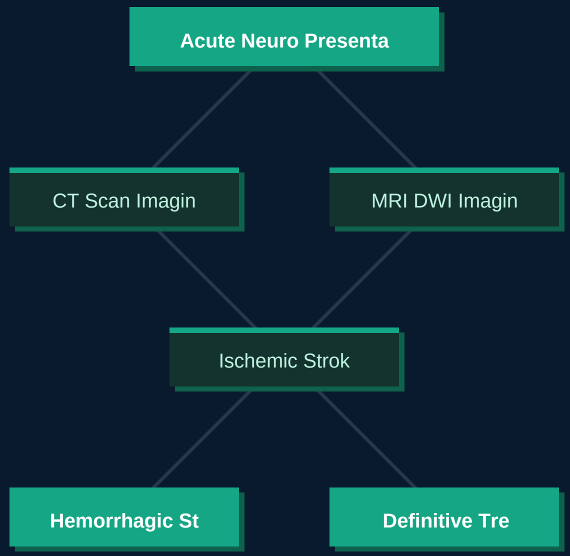
3. MRI with DWI detects ischemic stroke changes within minutes to hours of onset.

4. IV alteplase tPA is standard thrombolytic therapy within 4.5 hours of ischemic *stroke onset.*

5. Intracranial pressure monitoring indicated when GCS is 3 to 8 with abnormal CT *scan.*

6. Mechanical thrombectomy is effective up to 24 hours in selected large vessel *occlusion cases.*

CLINICAL PATHWAY



KEY INSIGHT

GCS below 8 mandates airway protection and ICP monitoring in neurotrauma.

Summary Mind Map: Human Brain

1 Brain weighs 1300 to 1400 g and consumes 20% of total body oxygen supply.

2 Four major divisions: cerebrum, cerebellum, brainstem, and diencephalon with *distinct roles.*

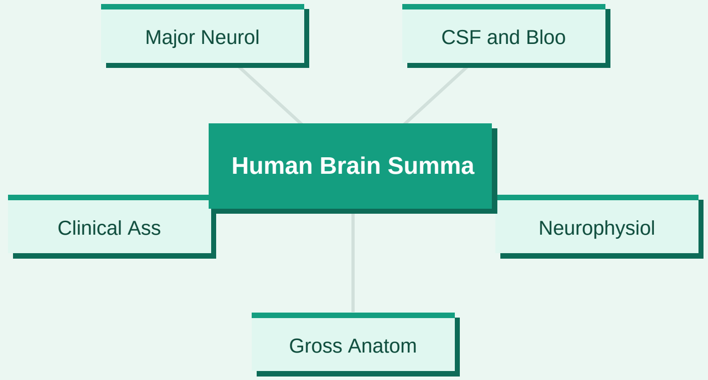
3 BBB formed by tight junctions; CSF produced at 500 mL per day by choroid *plexus.*

4 Key neurotransmitters: dopamine, serotonin, GABA, glutamate, and acetylcholine *in signaling.*

5 Major disorders include stroke, Alzheimer disease, epilepsy, Parkinson disease, *and TBI.*

6 GCS and neuroimaging guide acute management; tPA within 4.5 hours for *ischemic stroke.*

MIND MAP



KEY INSIGHT
Master anatomy, neurotransmitters, and imaging to excel in clinical neurology practice.

QUICK REFERENCE GUIDE

Key clinical terminology | SYNAPSE - aethex

Neuron
1

Basic excitable cell transmitting electrical signals via action potentials throughout the brain

Glia
2

Non-neuronal support cells including astrocytes, oligodendrocytes, and microglia in CNS

Blood-Brain Barrier
3

Selective endothelial barrier protecting brain from harmful blood-borne substances

Cerebrospinal Fluid
4

Clear fluid cushioning brain and spinal cord, produced by choroid plexus

Glasgow Coma Scale
5

15-point scale assessing eye, verbal, motor responses in brain injury patients

Substantia Nigra
6

Midbrain structure containing dopaminergic neurons lost in Parkinson disease

Corpus Callosum
7

White matter band connecting cerebral hemispheres with 200-250 million axons

Alteplase tPA
8

Thrombolytic drug used within 4.5 hours of ischemic stroke onset to dissolve clots